

Pedro Duarte

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Summary

Senior Software Engineer with 6 years of experience building scalable backend platforms in C# / .NET, microservices, and event-driven architectures with Kafka. Proven track record as a Tech Lead, owning system design, delivery, and production reliability across SQL and NoSQL data stores. Passionate about developer productivity and automation, including running a self-hosted Home Assistant environment with 50+ Zigbee devices and custom automations to experiment with distributed systems and real-world IoT integration.

Technical Skills

- **Backend:** C# / .NET 6–10, ASP.NET Core, Microservices architecture, REST APIs, Event-driven systems, Domain-driven design.
- **Data & Messaging:** Kafka, SQL Server, PostgreSQL, Elasticsearch, Redis, NoSQL data modelling, High-volume indexing & query optimization.
- **Infrastructure & DevOps:** Docker, Kubernetes, Azure DevOps Pipelines, Cloud-ready architectures, CI/CD migration (GoCD → Azure DevOps).
- **Architecture & Leadership:** Tech Lead experience, system design for scalable services, production reliability, security & vulnerability reduction to zero in production.
- **Automation & IoT (Personal Projects):** Self-hosted Home Assistant server, Zigbee network integration (50+ devices), automation design and monitoring.
- **Modern AI:** Practical use of GitHub Copilot and agentic coding tools to improve code quality, automate test creation, and speed up delivery of .NET microservices.

Professional Experience



Enhesa

(Current Company)

Senior Software Engineer - Tech Lead

Mar 2023 – Present

Tech stack: C# .NET 10, Kafka, Elasticsearch, PostgreSQL, Entity Framework, Kubernetes, ArgoCD, Redis, Structurizr

Responsible for the maintenance, design and development of core microservices of the company like Identity Service, User Management, Export Service, Document Service and Home Page.

I tackled a **major performance bottleneck in the Identity Service** where the /connect/token endpoint was averaging around 1.5 seconds per request under concurrent load, slowing down authentication across the platform. I redesigned the flow by introducing a distributed cache using Redis with TTL aligned to token expiration and added a circuit breaker to protect the system from cascading failures if Identity or cache dependencies became unstable. This reduced token retrieval latency from about 1.5 seconds to under 100 milliseconds, roughly a **93% improvement**, while significantly **reducing database load** and improving platform **resilience and scalability**.

In parallel, I addressed broader **platform scalability and delivery challenges** where integrations required code changes and release cycles, CI/CD pipelines were difficult to maintain, and several services carried production vulnerabilities. I designed and delivered an Integrations Service that enabled configuration-driven integrations without code changes, led a cross-service initiative that **reduced production vulnerabilities to zero**, and migrated pipelines **from GoCD to Azure DevOps Pipelines** to standardize and automate deployments. These changes reduced integration setup delivery time from weeks to hours or days, improved CI/CD reliability and strengthened security posture.



Enhesa

Software Engineer

May 2022 – Mar 2023

Tech stack: C# .NET, Kafka, SQL Server, Microservices



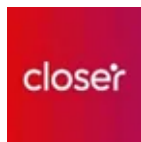
VORTAL

Software Engineer

Aug 2021 – Apr 2022

Tech stack: C# .NET Core, SQL Server, Entity Framework

Participated in the development of the initial microservices during the company's transition from a monolithic architecture to a microservice-based approach. I worked on building scalable backend services using C# .NET Core, SQL Server, and Entity Framework, contributing to the foundation of the new distributed platform and helping set patterns for future microservices.

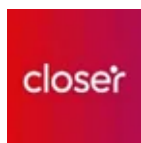


Closer Consulting

Junior Software Engineer

Sep 2020 – Aug 2021

Tech stack: .NET MVC, Razor, SQL Server



Closer Consulting

Intern Software Engineer

Jun 2020 – Sep 2020

Tech stack: .NET MVC, Razor, SQL Server

Education



**Polytechnic
Institute of Setúbal**

Bachelor's Degree - Management Information Systems

Sep 2017 – Jun 2020

Grade: 15/20

- Focused on the intersection of business and technology across the full information systems lifecycle
- Key coursework included application development, database management, systems analysis and design, architecture, project management, quality, security, and mathematics



Formabase

Certificate - Web Creation

Oct 2016 – Feb 2017

- Practical training in PHP, HTML, and CSS

Conferences and Events

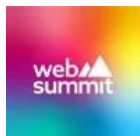


Azure Dev Summit

Participant

Oct 2025

Four-day conference focused on the Microsoft ecosystem, including a full-day hands-on workshop building an application with .NET Aspire.



Web Summit

Volunteer

Nov 2019

Volunteered in the days leading up to the event, helping with organization ahead of one of the world's best-known technology conferences.